Residual double-critical cubic fiber: constructive checkpoint

Executive conclusion

Fix distinct primes $q, r \geq 5$, and put

$$R_t = \mathbb{Z}/M_t\mathbb{Z}, \qquad M_t = q^{4t}r^{3t}, \qquad C_t = -3q^{2t}r^{2t}.$$

For

$$P_{C_t}(x,y) = (x-y)^3 + C_t(x+y), \qquad \mathcal{R}_t(A,B) = \frac{M_t^2|P_{C_t}(A,B)|}{|A||B|\max(|A|,|B|)},$$

the residual critical point does create an exponentially large unary fiber, but it does **not** create a Cartesian compression in any of the critical-cell, aligned-ball, finite multibranch, graph, or silent-CRT constructions that were tested rigorously.

The sharpest explicit critical construction found has a singleton image, yet

$$\mathcal{R}_t = q^{2\lceil t/2\rceil} \geq q^t;$$

it is exponentially worse than the cubic threshold. More strongly, if both inputs lie in one residual critical cell, then

$$\boxed{\mathcal{R}_t(A,B) \geq 1.}$$

For completely arbitrary $A, B \subseteq R_t$, the weighted product-tree argument gives the rigorous depth-dependent lower bound

$$\boxed{\mathcal{R}_t(A,B) \geq \frac{1}{60\mathcal{C}_{q,r}} q^{-t/2},} \tag{0.1}$$

where

$$\mathcal{C}_{q,r} = 6\left(A_q A_r^2 + A_q^2 A_r\right), \qquad A_p = (1-p^{-1})^{-1}.$$

No construction with $\mathcal{R}_t \to 0$ was found. However, (0.1) is not a uniform BP2+ estimate: an arbitrary recursively correlated family spread over exponentially many critical cells remains unresolved. Thus this checkpoint closes the **residual-fiber construction**, not the full arbitrary two-prime polarized problem.

No internet search or external verification was used.

A. Sub-agent assignments and independent conclusions

The common target given to all agents was to maximize the actual decay exponent

$$-\frac{1}{t}\log \mathcal{R}_t,$$

not collision energy or unary multiplicity.

1. **Critical q-adic geometry.** This branch computed the exact image and fibers of $u^3 - 3q^{2t}u \pmod{q^{4t}}$. The largest critical fiber is explicit and has $q^{\lfloor t/2\rfloor}$ residual pieces, but those pieces merely partition one flat ball.

2. **Cartesian realization.** This branch realized critical differences by sets $A \times B$, while charging $x + y$. The best singleton-image realization has $\mathcal{R}_t = q^{2\lceil t/2 \rceil}$; every pair of inputs in one critical cell has $\mathcal{R}_t \geq 1$.
3. **Correlated CRT encoding.** This branch used the r-coordinate to tag q-critical branches. A silent r-tag is exactly removable by saturation; non-silent transverse tags create full output cells. No graph or hierarchical code produced decay.
4. **Adversarial audit.** This branch searched for missing output digits and a family-wide floor. It found (0.1), and verified that the apparent $q^{t/2}$ gain is exactly the worst loss in the existing anchored-root estimate.
5. **Structured offline computation.** This branch tested balls, cell unions, digit masks, and CRT graphs at small parameters. It found no decreasing pattern. These experiments were used only to reject conjectures; no theorem below relies on them.

The assignments were non-overlapping: unary geometry, Cartesian realization, CRT correlation, adversarial proof, and structured experimentation were kept separate.

B. Best constructive candidate

Write

$$L = q^t r^t, \qquad C_t = -3L^2, \qquad M_t = q^t L^3, \qquad s = \left\lceil \frac{t}{2} \right\rceil.$$

Let ideals and cosets be taken in R_t, and define

$$\boxed{A_t = L + q^{t+s} r^t R_t, \qquad B_t = q^{2t} r^t R_t.} \tag{B.1}$$

Their exact sizes are

$$\boxed{|A_t| = q^{3t-s} r^{2t}, \qquad |B_t| = q^{2t} r^{2t}.} \tag{B.2}$$

This is the strongest direct use of the flat double root: it makes the entire Cartesian image a singleton. It nevertheless fails badly after the two input densities and the third mass factor are charged.

For comparison, the whole residual critical cell

$$H = LR_t$$

has

$$|H| = q^{3t} r^{2t}, \qquad |P_{C_t}(H, H)| = q^t, \qquad \mathcal{R}_t(H, H) = 1. \tag{B.3}$$

Thus the best proved asymptotic exponent among the surviving explicit Cartesian candidates is zero, not positive.

C. Exact critical-fiber calculation

First normalize away the q-adic unit contributed by r^t, and consider

$$f_t(u) = u^3 - 3q^{2t}u \pmod{q^{4t}}.$$

For $\varepsilon \in \{\pm 1\}$, $j \geq 1$, and

$$B_{\varepsilon,j} = \varepsilon q^t + q^{t+j}\mathbb{Z}/q^{4t}\mathbb{Z},$$

one has the exact expansion

$$f_t(\varepsilon q^t + q^{t+j}v) = -2\varepsilon q^{3t} + q^{3t+2j}v^2(3\varepsilon + q^j v). \tag{C.1}$$

Hence, before the modulus depth is reached,

$$v_q(f_t(\varepsilon q^t + q^{t+j}v) - f_t(\varepsilon q^t)) = 3t + 2j + 2v_q(v). \tag{C.2}$$

This is the promised quadratic flatness: there is no term linear in the perturbation.

The standard lifting fact used

For an odd prime p, every element of $1 + p\mathbb{Z}_p$ has a unique square root in $1 + p\mathbb{Z}_p$. Consequently, if a polynomial or convergent p-adic analytic map is congruent to the identity modulo p and has unit derivative, then it induces a bijection modulo every p^n. This is the ordinary unique-lifting form of Hensel's lemma.

Apply it to

$$v \longmapsto v\sqrt{1 + \frac{q^j v}{3\varepsilon}}.$$

If $n = t - 2j > 0$, (C.1) therefore gives the exact image

$$\boxed{f_t(B_{\varepsilon,j}) = -2\varepsilon q^{3t} + 3\varepsilon q^{3t+2j}\,\mathrm{Sq}(\mathbb{Z}/q^n\mathbb{Z}).} \tag{C.3}$$

If $n \le 0$, the image is the singleton $\{-2\varepsilon q^{3t}\}$.

The number of square residues modulo q^n is

$$S_q(n) = 1 + \frac{q-1}{2}\sum_{a=0}^{\lfloor (n-1)/2 \rfloor} q^{n-2a-1}, \tag{C.4}$$

or, equivalently,

$$S_q(n) = \begin{cases} 1 + \dfrac{q(q^n-1)}{2(q+1)}, & n \text{ even}, \\ 1 + \dfrac{q^{n+1}-1}{2(q+1)}, & n \text{ odd}. \end{cases} \tag{C.5}$$

Thus

$$|B_{\varepsilon,j}| = q^{3t-j}, \qquad |f_t(B_{\varepsilon,j})| = \begin{cases} S_q(t-2j), & 2j < t, \\ 1, & 2j \ge t. \end{cases} \tag{C.6}$$

The largest constant ball is

$$K_{\varepsilon,t} = \varepsilon q^t + q^{t+\lceil t/2 \rceil}\mathbb{Z}/q^{4t}\mathbb{Z}, \qquad |K_{\varepsilon,t}| = q^{\lfloor 5t/2 \rfloor}. \tag{C.7}$$

The complete critical fiber follows from

$$f_t(u) + 2q^{3t} = (u - q^t)^2(u + 2q^t)\,:$$

$$\boxed{f_t^{-1}(-2q^{3t}) = \left(q^t + q^{\lceil 3t/2 \rceil} R_q\right) \cup \left(-2q^t + q^{2t} R_q\right),} \tag{C.8}$$

where $R_q = \mathbb{Z}/q^{4t}\mathbb{Z}$. It has size

$$q^{\lfloor 5t/2 \rfloor} + q^{2t}. \tag{C.9}$$

The first ball in (C.8) splits into $q^{\lfloor t/2 \rfloor}$ balls of depth $2t$, all with the same output. They are not independent branches: they are exactly a partition of the one ball (C.7). The two critical signs have disjoint images because their centers differ by $4q^{3t}$, whereas every local variation is divisible by q^{3t+2}.

The unary ledgers expose the trap. For the largest critical fiber U_t,

$$\frac{q^{4t}|f_t(U_t)|}{|U_t|^2} \asymp_q q^{-t}, \qquad \frac{q^{8t}|f_t(U_t)|}{|U_t|^3} \asymp_q q^{t/2}. \tag{C.10}$$

The first quantity looks constructively attractive; the second is the actual cubic/BP ledger and loses.

D. Exact image-size estimate for the Cartesian candidate

Take $x = L + a \in A_t$ and $y = b \in B_t$, and put $h = a - b$. Since

$$P_{C_t}(x, y) = f(h + L) + 2C_t b, \qquad f(u) = u^3 - 3L^2 u,$$

the critical expansion is

$$f(L + h) - f(L) = 3Lh^2 + h^3. \tag{D.1}$$

Here

$$v_q(h) \geq t + s, \qquad v_r(h) \geq t.$$

Therefore

$$v_q(3Lh^2) \geq 3t + 2s \geq 4t, \qquad v_r(3Lh^2) \geq 3t,$$

and the cubic term is at least as divisible. Also

$$2C_t b \equiv 0 \pmod{q^{4t} r^{3t}}.$$

Since $f(L) = -2L^3$, this proves the exact, not energetic, estimate

$$\boxed{P_{C_t}(A_t, B_t) = \{-2L^3\}, \qquad |P_{C_t}(A_t, B_t)| = 1.} \tag{D.2}$$

Why all restrictions inside one critical cell fail

Let

$$H = LR_t,$$

fix a coset $c + H$, and take arbitrary $A, B \subseteq c + H$. Write

$$x = c + L\xi, \qquad y = c + L\eta.$$

Then

$$P_{C_t}(x, y) = -6L^2c + L^3((\xi - \eta)^3 - 3(\xi + \eta)). \tag{D.3}$$

The bracket matters only modulo q^t. Define

$$\pi(c + L\xi) = \xi \pmod{q^t}, \qquad X = \pi(A), \quad Y = \pi(B).$$

Each π-fiber has $L^2 = q^{2t}r^{2t}$ elements, and multiplication by L^3 injects $\mathbb{Z}/q^t\mathbb{Z}$ into R_t. Hence

$$\boxed{|P_{C_t}(A, B)| = |P_{-3}(X, Y)|, \qquad |A| \le L^2|X|, \qquad |B| \le L^2|Y|.} \tag{D.4}$$

For the unit-coefficient map P_{-3} modulo q^t, the change $(x, y) \leftrightarrow (u = x - y, v = x + y)$ is invertible. Once an output and u are fixed, v is unique because -3 is a unit. Thus every full-ring fiber has exactly q^t points, and

$$|P_{-3}(X, Y)| \ge \frac{|X||Y|}{q^t}. \tag{D.5}$$

Combining (D.4) and (D.5) yields

$$\boxed{\mathcal{R}_t(A, B) \ge \frac{q^t}{\max(|X|, |Y|)} \ge 1.} \tag{D.6}$$

This is a complete no-go theorem for arbitrary sparse digits, progressions, Cantor restrictions, and unions of fine balls contained in one residual critical cell.

A universal bound for arbitrary cells

The following proposition records exactly what remains of the old weighted argument after the residual double root appears.

Proposition. For all nonempty $A, B \subseteq R_t$,

$$|P_{C_t}(A, B)| \ge \frac{q^{-t/2}}{60\mathcal{C}_{q,r}} \frac{|A||B| \max(|A|, |B|)}{M_t^2}. \tag{D.7}$$

Proof. Invoke the arbitrary-exponent form of the exact product-tree difference estimate from the supplied sum-product construction. It holds for $M = q^\ell r^m$, and hence for $(\ell, m) = (4t, 3t)$:

$$\sum_{x \in A, y \in B} \left(\frac{\mathrm{ord}(x - y)}{M_t} \right)^2 \ge \frac{|A|^2|B|^2}{4\mathcal{C}_{q,r}M_t^2}. \tag{D.8}$$

Here $\mathrm{ord}(u)$ is the additive order of u, with $\mathrm{ord}(0) = 1$, and we set $v_p(0) = +\infty$. Define

$$\Omega_t(u) = q^{-2\min(v_q(u),t)} r^{-2\min(v_r(u),t)}.$$

Because this truncation is shallower than the exponents in M_t, $\Omega_t(u) \geq (\mathrm{ord}(u)/M_t)^2$. Therefore

$$\sum_{A\times B} \Omega_t(x-y) \geq \sum_{A\times B} \left(\frac{\mathrm{ord}(x-y)}{M_t}\right)^2 \geq \frac{|A|^2|B|^2}{4\mathcal{C}_{q,r} M_t^2}. \tag{D.8a}$$

For the r-adic anchored cubic modulo r^{3t}, the frozen exact 2:3 root calculation gives weighted root mass at most 3, for either sign of the linear term. At the q-place, split roots of

$$u^3 \pm 3q^{2t}c^2 u = w \pmod{q^{4t}}, \qquad q \nmid c,$$

according to $h = v_q(u)$:

1. If $h < t$, the cubic term dominates. There are at most $3q^{2h}$ lifted roots, and their total weight is at most 3.
2. If $h > t$, the linear term is transverse after scaling. The discarded q^{2t} lifts cancel the weight q^{-2t}, contributing at most 1.
3. If $h = t$, divide by q^{3t}. One obtains $v^3 \pm 3c^2 v = a \pmod{q^t}$. If critical residue classes exist, their critical values are distinct, so at most one critical class can occur for a fixed a. In that class, the same analytic change used in (C.3) reduces the equation to a square congruence, which has at most $2q^{\lfloor t/2 \rfloor}$ roots. If a critical class occurs, there is at most one other simple class; if no critical class occurs, there can be at most three simple classes. Every solution $v \pmod{q^t}$ has q^{2t} lifts in the original scaled variable, and the factor q^{-2t} in Ω_t cancels those lifts.

Consequently the q-adic weighted root mass is at most

$$2q^{\lfloor t/2 \rfloor} + 5 \leq 5q^{t/2}.$$

CRT makes the two local sums multiply, so every anchored global root fiber has weighted mass at most $15q^{t/2}$. Fixing y gives

$$u^3 + C_t u = z - 2C_t y,$$

while fixing x gives

$$u^3 - C_t u = z - 2C_t x.$$

The preceding estimate covers both signs. Anchoring in these two ways shows that every output z satisfies

$$\sum_{\substack{(x,y)\in A\times B \\ P_{C_t}(x,y)=z}} \Omega_t(x-y) \leq 15q^{t/2}\min(|A|,|B|). \tag{D.9}$$

Group the total weighted pair mass by outputs, combine (D.8) and (D.9), and use

$$\frac{|A|^2|B|^2}{\min(|A|,|B|)} = |A||B|\max(|A|,|B|).$$

This proves (D.7). □

The factor $q^{t/2}$ in (D.9) is precisely the residual critical multiplicity. The proposition proves that no construction in this family can decay faster than $q^{-t/2}$, but it does not rule out decay at that scale.

E. Normalized compression ledger

For the singleton candidate (B.1), $|A_t| \geq |B_t|$, so (B.2) and (D.2) give

$$\boxed{\mathcal{R}_t(A_t, B_t) = \frac{q^{8t} r^{6t}}{(q^{3t-s} r^{2t})^2 (q^{2t} r^{2t})} = q^{2s} = q^{2\lceil t/2 \rceil}.} \tag{E.1}$$

Thus

$$-\frac{1}{t} \log \mathcal{R}_t(A_t, B_t) = -\frac{2\lceil t/2 \rceil}{t} \log q \longrightarrow -\log q.$$

It loses exponentially. The main candidates rank as follows.

1. The unary full critical fiber has $Q_2 \asymp q^{-t}$, but its cubic ledger is $Q_3 \asymp q^{t/2}$; it is not Cartesian.
2. The singleton Cartesian pair (B.1) has $\mathcal{R}_t = q^{2\lceil t/2 \rceil}$ and decay exponent $-\log q + o(1)$.
3. The whole critical cell (H, H) has $\mathcal{R}_t = 1$, hence exponent zero.
4. Every pair inside one common H-cell has $\mathcal{R}_t \geq 1$.
5. An arbitrary pair in the full ring obeys $\mathcal{R}_t \geq (60\mathcal{C}_{q,r})^{-1} q^{-t/2}$, which caps any possible decay exponent at $\frac{1}{2} \log q + o(1)$.

No explicit candidate has a positive decay exponent.

F. Same-set conversion

The asymmetric singleton construction does not survive union into one set. Let

$$S_t = A_t \cup B_t.$$

Since $|S_t| \leq 2|A_t|$ and

$$P_{C_t}(x, x) = 2C_t x,$$

the diagonal block gives

$$|P_{C_t}(S_t, S_t)| \geq |2C_t A_t| = q^{t-s}.$$

Consequently

$$\boxed{\frac{M_t^2 |P_{C_t}(S_t, S_t)|}{|S_t|^3} \geq \frac{1}{8} q^{2s} \geq \frac{1}{8} q^t.} \tag{F.1}$$

There is also a general same-set obstruction. Multiplication by $2C_t$ has kernel size

$$g = \gcd(C_t, M_t) = L^2.$$

Therefore every $S \subseteq R_t$ satisfies

$$|P_{C_t}(S, S)| \geq \frac{|S|}{L^2},$$

and hence

$$\boxed{\mathcal{R}_t(S,S) \geq \left(\frac{|H|}{|S|}\right)^2.} \tag{F.2}$$

If S occupies at most J cosets of H, then $\mathcal{R}_t(S,S) \geq J^{-2}$. Any exponentially compressive same set would therefore have to occupy exponentially many critical cells and control all off-diagonal cross-pairs.

Even perfect control of four blocks at a very sparse scale does not help. Put

$$T = q^{2t} r^t R_t, \qquad S = T \cup (L + T).$$

The four blocks have exactly the outputs

$$0, \qquad -2L^3, \qquad -4L^3, \qquad -6L^3,$$

so $|S| = 2L^2$ and $|P_{C_t}(S,S)| = 4$, but

$$\mathcal{R}_t(S,S) = \frac{1}{2} q^{2t}. \tag{F.3}$$

G. Global amplification test

No exponential local compression was proved. The best explicit local gain parameter is therefore

$$\eta = 0.$$

Accordingly, an amplification condition

$$\eta\theta > \kappa_{\text{global}}$$

cannot hold for any positive archimedean/root-discriminant cost κ_{global}. There is no justified number-field ledger and no candidate real exponent $c > 0$ to derive from this family.

The lower bound (D.7) says only that a hypothetical construction could gain at most $\frac{1}{2}\log q$ per unit of t. A lower bound is not a construction and cannot be entered on the gain side of the global ledger.

H. Verdict toward the ultimate theorem

Selected verdict: 3. No constructive gain.

The rigorous scope of that verdict is important:

The residual double-critical fiber mechanism is closed, but full BP2+ for arbitrary cross-cell A, B is not yet proved.

Thus "BP2+ survives" here means that it remains unrefuted and is proved on the entire critical cell plus all tested structured families. It is **not** a claim of a uniform family-wide theorem. Closing the entire cubic local stream would currently overstate the mathematics.

The exact missing configuration is a recursively nontransverse, sparse prefix-code spread over a growing number of H-cells. Mass-only tensorization does not control it, and none of the present constructions makes it compress.

I. Exactly one next mathematical target

Prove or disprove the following full-cell translated charging statement.

Let $H = q^t r^t R_t$, let $K = |H| = q^{3t} r^{2t}$, and let $X, Y \subseteq R_t/H$. Is there a constant $b_{q,r} > 0$, independent of t, X, Y, such that

$$\boxed{|P_{C_t}(X + H, Y + H)| \geq b_{q,r} K \frac{|X||Y| \max(|X|, |Y|)}{(q^t r^t)^2}?} \tag{I.1}$$

This removes all within-cell sparsity while retaining the unresolved translation and cross-cell overlap. It is the smallest clean statement that tests whether exponentially many residual-cell addresses can cooperate.

Frozen-input and theorem-use ledger

The exact 2:3 no-compression theorem and its anchored weighted root estimate were treated as frozen and were not re-proved. The only substantial general tools invoked above were:

1. the Chinese remainder theorem;
2. the unique-lifting form of Hensel's lemma stated in Section C;
3. the exact product-tree difference inequality (D.8) from the supplied sum-product construction.

All remaining arguments are explicit congruence, valuation, fiber, and cardinality calculations contained in this note.